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A Recurrent Neural Network based deep learning model for offline signature verification and recognition system

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ABSTRACT

With the recent advancement in information technology field, the demand to develop a person authentication system through verifying their offline signatures is gradually increasing. This type of system may be used to verify various official documents through verifying the signatures of the concerned persons present in the documents. This article proposes a Recurrent Neural Network (RNN), a deep learning network, based method to verify and recognize offline signatures of different persons. Various structural and directional features have been extracted locally from each signature sample and the generated feature vectors have been studied using two different models of RNN—long-short term memory (LSTM) and bidirectional long–short term memory (BLSTM). The performance of the proposed system has been tested on six widely used public signature databases—GPDS synthetic, GPDS-300, MCYT-75, CEDAR, BHSig260 Hindi, and BHSig260 Bengali. Experiment has also been performed using Convolutional Neural Network (CNN) to have a comparison with RNN based results. Experimental results demonstrate that the proposed RNN based signature verification and recognition system is superior over CNN and also outperforms the existing state-of-the-art results in this regard.

1. Introduction

Signature, one of the popular biometric methods for person authentication, represents the behavioural features of an individual instead of physical features such as fingerprint, iris, face, etc. Signatures are widely used to verify various banking, legal, business, insurance documents, among others, through authenticating the persons put their signatures on those documents. Therefore, the development of any automated signature verification and recognition system will speed up the verification process of these types of documents. Handwritten signature verification and recognition can be performed in two different modes—offline and online. In offline signature acquisition, people put their signatures on a piece of paper and these signatures are eventually recognized and verified by the system after feeding the scanned copy of the paper to the system in image form, whereas in online mode signatures are put directly on the computer monitor using some digitizing tablet and system recognizes and verifies those immediately. Accurate verification and recognition of signatures in offline mode are more difficult than in online mode due to absence of dynamic information of signature images as present in online mode (Ghosh, Kumar, & Roy, 2019a). These systems are usually divided into two sub-systems; signature verification (SV) and signature recognition (SR). In signature verification, features of the test signature are compared with the features of a set of reference signatures whose identity is claimed (Ghosh et al., 2019a), whereas in signature recognition the system recognizes any individual's genuine signature and predicts the identity of the person (García, Ramos-Lara, Miguel-Hurtado, & Cantó-Navarro, 2014; Ghosh et al., 2019a). Basically, signature verification system verifies whether a signature is genuine or forged and if it is genuine, then the signature recognition system reveals the identity of the person producing the genuine signature. So, both genuine and forged signatures are tested by signature verification system, whereas signature recognition is performed always on genuine signatures written by authentic writers. In random forgery (Malik & Liwicki, 2012), genuine signature of any person is produced by another unauthentic person, whereas in skilled forgery (Malik & Liwicki, 2012), experienced forgers produce genuine signatures of authentic persons after practising for a long time. So, the forged signatures produced through random or skilled forgeries as well as the genuine signatures are verified by the signature verification system to check whether those are genuine or forged. Again, signature verification methods can be divided into writer dependent (WD) and writer independent (WI) modes. In WD mode, a separate classifier is created for each signer and the classifier is retrained each time a new signer is enrolled to the system (Bhunia, Alaei, & Roy, 2019; Bouamra, Djeddi, Nini, Diaz, & Siddiqi, 2018; Hafemann, Sabourin, & Oliveira, 2016), whereas in WI mode a generic system is developed that can test the signature of a new signer without retraining the classifier (Dutta, Pal, & Llados, 2016; Kumar, Sharma, & Chanda, 2012; Zois, Alexandridis, & Economou, 2019a).

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Several investigation reports (Bhunia et al., 2019; Jagtap, Sawat, Hegadi, & Hegadi, 2020; Kumar et al., 2012; Vargas, Ferrer, Travieso, & Alonso, 2011) are available on the development of offline signature verification and recognition systems and these studies have tried to improve the overall accuracy of verification and recognition. Most of these studies have used various shallow machine learning (ML) techniques such as Siamese Neural Network (SNN) (Jagtap et al., 2020), Support Vector Machine (SVM) (Bhunia et al., 2019; Diaz, Ferrer, Eskander, & Sabourin, 2017; Ferrer, Vargas, Morales, & Ordonez, 2012; Guerbai, Chibani, & Hadjadji, 2015; Hafemann, Oliveira, & Sabourin, 2018; Kumar et al., 2012; Okawa, 2018a; Pal, Pal, & Blumenstein, 2013), shallow Convolutional Neural Network (CNN) (Jain, Singh, & Singh, 2020), K-Nearest Neighbour (KNN) classifier (Hezil, Djemili, & Bourouba, 2018; Pal, Alaei, Pal, & Blumenstein, 2016), Multilayer Perceptron (MLP) (Drouhard, Sabourin, & Godbout, 1996; Kumar et al., 2012; Pal, Alaei, Pal, & Blumenstein, 2015; Shekar, Bharathi, Kittler, Vizilter, & Mestestskiy, 2015), Hidden Markov Model (HMM) (Justino, Yacoubi, Bortolozzi, & Sabourin, 2000; Wen, Fang, Tang, & Zhang, 2009; Yilmaz, Yanikog, Tang, & Zhang, 2016), Bayes classifier (Kalera, Srihari, & Xu, 2004), etc. Grid based template matching scheme has also been used (Zois, Alewijnse, & Economou, 2016) for offline signature verification. Deep CNN based exploration has also been reported in the literature (Hafemann et al., 2018; Hafemann, Sabourin, & Oliveira, 2017). But, no study has been found in the literature in this domain employing another deep learning technique, Recurrent Neural Network (RNN). RNN is basically used for sequence labelling task. RNN has two different models—long-short term memory (LSTM) (Ghosh, Vamshi, & Kumar, 2019b) and bidirectional long-short term memory (BLSTM) (Ghosh et al., 2019b). RNN, through LSTM and BLSTM models, can remember a long sequence of symbols over a longer period of time. In addition to this, BLSTM can remember any sequence in both forward and backward directions. Offline signatures are written as a sequence of few known characters in any script or as a combination of few unknown trajectories. By exploiting the sequence memorizing capability of LSTM and BLSTM models of RNN any system can recognize any character or trajectory in a signature by knowing the characters or trajectories appearing both before and after it. This capability is not present in any of the shallow machine learning techniques. Due to this capability, deep learning model using RNN produces better results (Section 5 may be referred) on any sequential data related problem in comparison to shallow machine learning models. The present article proposes a RNN based deep learning model to verify and recognize offline handwritten signatures in WD mode. The experiments have also been carried out using CNN to have a comparison with RNN based results.

The rest of the paper is organized as follows: Related works are discussed in Section 2. Description of different datasets used in the present work is given in Section 3. The details of the proposed method are presented in Section 4. Section 5 analyses the verification and recognition results of the present system. Finally, Section 6 concludes the paper with a direction for future research.

2. Literature survey

Several research investigation reports are available in the literature in various scripts on offline handwritten signature verification and recognition systems. Jagtap et al. (2020) presented one report on verification of genuine and forged offline signatures using SNN. CNN has been used as a subnetwork in this investigation. SVM has been used extensively for signature verification purpose. Bhunia et al. (2019) presented one SVM based WD method for signature verification by combining the hybrid texture features. Kumar et al. (2012) proposed a set of features for offline WI signature verification system by exploiting the surroundedness property of the signature. The feature vectors were studied using two different classifiers—SVM and MLP. Hafemann et al. (2016) proposed a WD training scheme using SVM classifier after learning the features using deep CNN in WI format for offline signature verification system. The co-occurrence matrix and local binary pattern were also used as features in the literature (Vargas et al., 2011). The signature verification was performed by SVM in this study. Pal et al. (2013) proposed another SVM based method for offline signature verification in Devanagari script. The gradient and Zernike moment features were extracted from offline signature images. The problem of robustness of grey level features of signature strokes when the signature is written on a complex background was dealt in another study (Ferrer et al., 2012). Guerbai et al. (2015) presented an one-class SVM classifier based offline signature verification system using WI parameters. This study considered genuine signatures only in the training set. Another one-class SVM based WD signature verification system was reported by Bouamra et al. (2018). Hafemann et al. (2018) presented a method using deep CNN features for offline signature verification, where the classification was performed using SVM classifier in WD mode. The study tried to solve the problem of feeding fixed-sized signature representation to the input layer of the CNN by generating a fixed-sized representation of variable-sized signatures. Another study (Diaz et al., 2017) proposed textural features based signature verification method using SVM classifier. Okawa (2018a) reported a WD signature verification system where features generated from Fisher vector were fused with KAZE features. The verification was performed using SVM classifier. Dutta et al. (2016) proposed a combination of local features and their global statistics for WI offline signature verification. The local features were paired using the edges of a graph. The classification was performed using SVM. Few shallow CNN based offline signature verification systems are also available in the literature as well. Jain et al. (2020) proposed a shallow CNN based language-independent method for offline signature verification and the method performance was tested on signatures of three different scripts—Latin, Devanagari, and Bengali. In another study (Hezil et al., 2018), the method relied on KNN classifier for recognition of offline signatures. Features were extracted using statistical image feature and local binary pattern descriptors. KNN classifier was used for classification purpose in another signature verification system (Pal et al., 2016) also where texture based features were fed to the classifier. Drouhard et al. (1996) presented one MLP based method for offline signature verification by exploiting the directional Probability Density Function (PDF). Pal et al. (2015) presented one Probabilistic Neural Network (PNN) based method for offline signature verification. Signatures were characterized using connected components, enclosed regions, and curvelet features. A grid structured morphological pattern based method was proposed in Shekar et al. (2015) where signature verification was performed using SVM and MLP. Before extracting the features, the entire signature image was divided into eight equally-sized vertical grids. A hybrid method of offline signature verification using Discrete Radon Transform (DRT) and PNN was proposed in Ooi, Teoh, Pang, and Hiew (2016). Wen et al. (2009) presented one HMM based offline signature verification method employing rotation invariant structure features. In this study, discrete fast Fourier transform was used to remove the phase shift. Score level fusion of classifiers using HMM for offline signature verification was proposed in Yilmaz et al. (2016). Both WD and WI classifiers were investigated in this study. Justino et al. (2000) presented a HMM based offline signature verification system using graphometric features. In another investigation report (Kalera et al., 2004), a Bayes classifier based offline signature verification method was presented using gradient, structural, and concavity features. In another study (Serdouk, Nemmour, & Chibani, 2016) on offline signature verification system, gradient and topological features were proposed to develop Artificial Immune Recognition System (AIRS). The AIRS was trained using WD approach. Grid based template matching scheme for WD offline signature verification has also been reported in the literature (Zois et al., 2016). Features were extracted from lattice shaped structures of 5×5 pixel window size. Another notable study (Hafemann et al., 2017) addressed the problem of unavailability of dynamic information from signature images in offline mode and proposed a deep CNN based

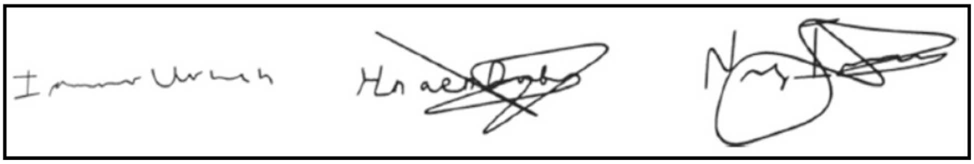

Fig. 1. Few signature samples from GPDS synthetic dataset.

method to learn the representations directly from the signature images in WI format. In this study, the features of the forged signatures were learnt by the classifier during the training phase. The classifiers were trained in WD approach. Fuzzy modelling based similarity measurement schemes (Ferrer, Alonso, & Travieso, 2005; Hanmandlu, Yusof, & Madasu, 2005) are also available in the literature for offline signature verification. Hanmandlu et al. (2005) presented a fuzzy method utilizing the Takagi–Sugeno model. In another study Ferrer et al. (2005) proposed a set of geometric signature features under fuzzy modelling environment. A signature identification method (Hadjadji, Chibani, & Nemmour, 2017) based on Curvelet Transform (CT) and Principal Component Analysis (PCA) has also been reported in the literature. CT was utilized for feature generation and PCA was explored to reduce the dimensionality of feature vectors. A WI offline signature verification study using asymmetric pixel relations and unrelated training-testing datasets is also available in the literature (Zois et al., 2019a). In another study (Okawa, 2018b), Okawa proposed an offline signature verification method using feature encoding scheme of a bag-of-visual words. The study considered the current cognitive knowledge used by the forensic document examiners. In another notable study (Soleimani, Araabi, & Fouladi, 2016), a new classification method known as Deep Multitask Metric Learning (DMML) was proposed for offline signature verification. In a recent research effort on offline signature verification (Maergner, et al., 2019), a combination of graph edit distance with deep triplet networks has been proposed. The power of sparse representation has been exploited to extract features in another recent effort (Zois, Tsourounis, Theodorakopoulos, Kesidis, & Economou, 2019b) on offline signature verification. The use of image visibility graphs to map images into networks was introduced for WD offline signature verification in another recent study (Zois, Zervas, Tsourounis, & Economou, 2020). The survey shows that no investigation report using RNN based deep learning network has been reported in this domain in the literature.

3. Dataset description and experimental protocol

The present investigation has used GPDS synthetic (Ferrer, Diaz-Cabrera, & Morales, 2013, 2015), GPDS-300 (Ferrer et al., 2013, 2015), MCYT-75 (Aguilar, et al., 2005; Ortega-Garcia et al., 2003), and CEDAR public signature databases (Bharathi & Shekar, 2013) for signature verification and recognition. The signatures were written in Latin script in these three datasets. To show the pertinency of the proposed method on other scripts, the performance of the present system has been tested on BHSig260 Hindi (Pal et al., 2016) and BHSig260 Bengali (Pal et al., 2016) datasets also. BHSig260 Hindi and BHSig260 Bengali datasets contain handwritten signatures in Devanagari and Bengali scripts respectively. The details of the datasets used in the present investigation are elaborated below.

3.0.1. GPDS

GPDS synthetic[1] dataset contains offline signatures of 4000 individuals. GPDS synthetic dataset has various versions. The present work has used GPDS synthetic as well as GPDS-300 versions. The GPDS-300 version contains signatures of first 300 individuals out of 4000. This dataset contains both genuine and forged signatures. 24 genuine and 30 forged samples were collected for each individual signature, leading to a total of 96000 genuine and 120000 forged signature samples in GPDS synthetic dataset and 7200 genuine and 900 forged signature samples in GPDS-300 version. As signatures were collected in offline mode, so dynamic information like pen pressure, pen speed, etc., generated during online mode of writing are not present in this dataset (Ferrer et al., 2013, 2015).

Fig. 3. Few signature samples from CEDAR dataset.

3.0.2. MCYT-75

MCYT-75[2] dataset contains offline handwritten signatures of 75 individuals. 15 genuine and 15 forged samples were collected for each individual signature (Aguilar, et al., 2005; Ortega-Garcia et al., 2003). So, a total of 1125 genuine and 1125 forged signature samples are available in MCYT-75 dataset.

3.0.3. CEDAR

Centre of Excellence for Document Analysis and Recognition (CEDAR) signature dataset contains signatures of 55 individuals (Bharathi & Shekar, 2013). For each individual signature, 24 genuine and 24 forged samples were collected, leading to a total of 1320 genuine and 1320 forged signature samples.

3.0.4. BHSig260

BHSig260 signature dataset[3] has two different versions—BHSig260 Hindi and BHSig260 Bengali. In the BHSig260 Hindi version, 160 individuals have provided their signatures in Devanagari script. For each individual signature, 24 genuine and 30 forged samples were collected, leading to a total of 3840 genuine and 4800 forged signatures samples. In the BHSig260 Bengali version, there were a total of 100 signers and 24 genuine and 30 forged samples were collected for each individual signature in Bengali script (Pal et al., 2016). So, a total 2400 genuine and 3000 forged signature samples are present in this version.

Each signature dataset has been divided into training and test sets. The classifier has been trained using genuine signature samples only of all the signers and the number of training samples is same for each individual genuine signature in each dataset. The test set contains all the forged signature samples and the remaining genuine signature samples not present in the training set. The detailed statistics of different datasets is shown in Table 1. The experimental protocol has been repeated by selecting different sets of genuine signature samples (but equal number of samples for each individual genuine signature as mentioned in Table 1) in the training set and accordingly different sets of genuine signature samples in the test set. Few samples from all the aforesaid datasets are shown in Figs. 1–5.

[1] http://www.gpds.ulpgc.es/downloadnew/download.htm

[2] http://atvs.ii.uam.es/atvs/mcyt75so.html

[3] https://drive.google.com/file/d/0B29vNACcjvzVc1RfVkg5dUh2b1E/view

Table 1
Details of the Training and Testing Signature Datasets.

Script	Dataset	No. of Writers	Training samples per individual signature	Total No. of training samples	Total No. of test samples
Latin	GPDS synthetic	4000	12	48 000 genuine	48 000 genuine, 120 000 forged
	GPDS-300	300	12	3600 genuine	3600 genuine, 9000 forged
	MCYT-75	75	10	750 genuine	375 genuine, 1125 forged
	CEDAR	55	12	660 genuine	660 genuine, 1320 forged
Devanagari	BHSig260 Hindi	160	12	1920 genuine	1920 genuine, 4800 forged
Bengali	BHSig260 Bengali	100	12	1200 genuine	1200 genuine, 3000 forged

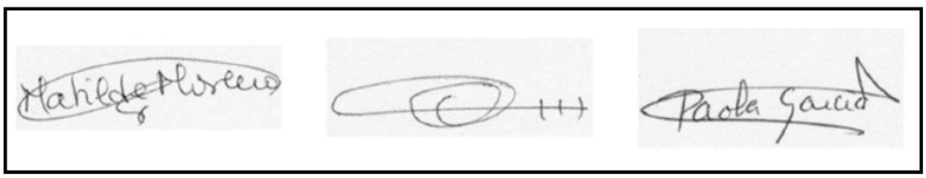

Fig. 2. Few signature samples from MCYT-75 dataset.

4. Proposed methodology

In the proposed method, the collected raw genuine signature samples in different training sets have been preprocessed to remove the variations within the samples of same signature. After preprocessing, various structural and directional features have been extracted from the signature trajectories to discriminate one signature from another. The LSTM and BLSTM models of RNN classifier have then been trained using the generated feature vectors of genuine signature samples. In the present system, signatures are verified first and if those are genuine, then the system goes for recognizing those signatures. For the verification purpose, both genuine and forged signatures have been used, whereas for the recognition purpose only genuine signatures have been utilized. A flowchart of the proposed SV and SR method is shown in Fig. 6.

The proposed method is described below in details:

4.1. Preprocessing

It is a well known fact that intra-class variation occurs in signature samples. The variation among the samples of the same individual signature occurs due to the variation in human mood, mental state, etc. of a person as well as lack of space on the writing surface. Due to these factors, some samples of the same signature often vary in height, width, skew nature, etc. To remove these intra-class variations, a number of preprocessing steps (resizing and skew correction) have been performed in the present investigation. The images of all the signature samples have been resized to a standard 128 × 128 pixels to remove the height and width variations among the samples of the same signature. If the height and width vary among the samples of the same individual signature then similar feature values may not be obtained from the samples of the same signature as few of the octants may remain vacant during extracting various features and naturally dissimilar feature values will be obtained from the samples of the same signature (Section 4.2 may be seen for the methods of extracting various features). After resizing, the samples of the same individual signature that are skewed, not horizontally oriented, have been brought into horizontal orientation by correcting the skew. The details of the skew correction method may be seen in Ghosh et al. (2019b). If some samples of the same individual signature are skewed and remaining are horizontally oriented, then also dissimilar feature values will be obtained from the samples of the same individual signature as the distribution of the signature pixels to different octants will not be uniform even for the samples of the same individual signature. This issue is illustrated in Fig. 7.

4.2. Feature extraction

Feature extraction phase follows the preprocessing phase. In this phase, various characteristics or features able to discriminate different signatures have been extracted from each signature sample. In the present investigation, four structure and direction oriented features namely, change of trajectory direction, trajectory slope, trajectory waviness, and centre-of-mass in varying numbers have been extracted from each signature sample. The methods of extracting the aforesaid features are discussed below in details.

a. Change of trajectory direction (CTD): The entire signature image is partitioned into 8 octants and the change of direction of the signature portion (consists of black pixels) falls in each octant is computed. A square box of fixed area enclosing the entire signature image is considered for octant division of all the signature images. The feature values are extracted sequentially octant wise in the order from octant 1 to octant 8. To compute this direction, the first and last black pixels as well as the centroid of all the black pixels in each octant are considered. The first and last black pixels in each octant have been decided in accordance with the x-coordinate value i.e., the black pixel with the least x-coordinate value within any octant is designated as the first pixel and the one with the highest x-coordinate value within any octant is designated as the last black pixel. Then two straight lines; one connecting the first black pixel and the centroid and the other connecting the centroid and the last black pixel, are formed which create two angles (β_1 and β_2) with the x-axis (shown in Fig. 8). Cos(β_1), Sin(β_1), Cos(β_2), and Sin(β_2) are considered as the change of trajectory direction in each octant. Here, the Cos and Sin of different angles provide normalized feature values in the range −1 to +1. If any of the octants does not contain any black pixel then these four feature values have been considered as “0” in that octant. Thus, a total of 32 feature values are obtained from this feature. Fig. 8 illustrates the

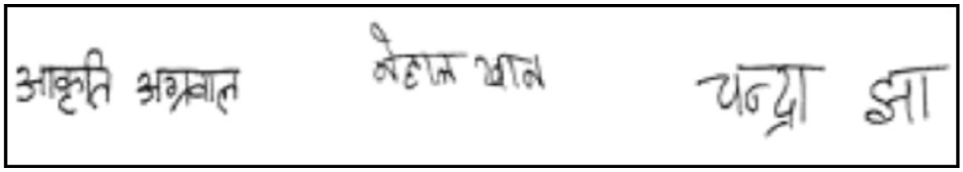

Fig. 4. Few signature samples from BHSig260 Hindi dataset.

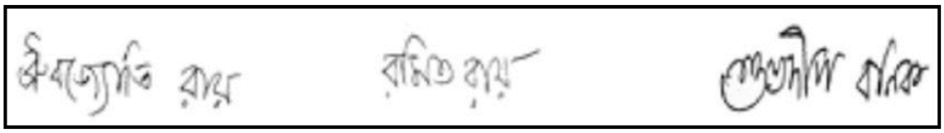

Fig. 5. Few signature samples from BHSig260 Bengali dataset.

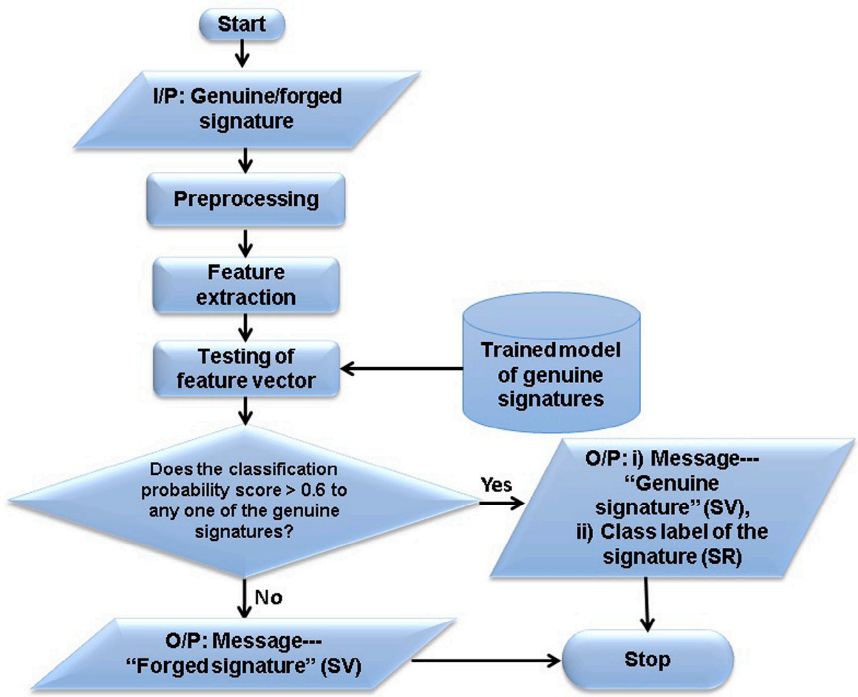

Fig. 6. Flowchart of the proposed SV and SR method.

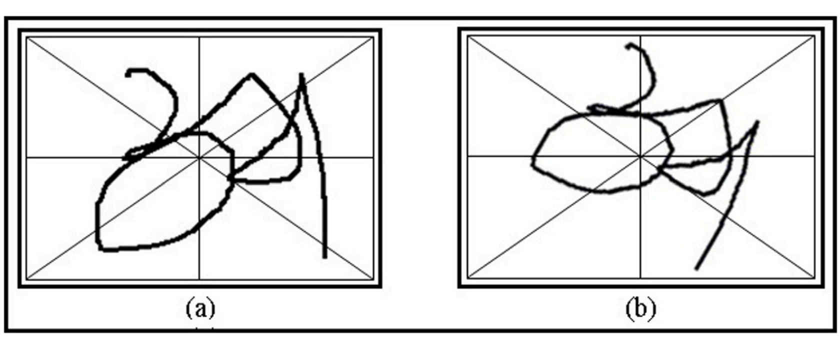

Fig. 7. Illustration of different orientations of two samples of the same signature in Devanagari script: (a) Horizontally oriented, (b) Skewed.

idea of extracting this feature from one signature sample in Devanagari script.

b. Trajectory slope (TS): Before extracting this feature also, the entire signature image is partitioned into 8 octants and the slope of the signature portion falls in each octant is computed. In this method of feature extraction also, a square box of fixed area enclosing the entire signature image is considered for octant division of all the signature images. The feature values are extracted sequentially octant wise in the order from octant 1 to octant 8. To compute this feature, only the first and last black pixels in each octant are considered. The first and last black pixels in each octant have been decided in the same way as discussed in feature (a). Then a straight line, connecting these

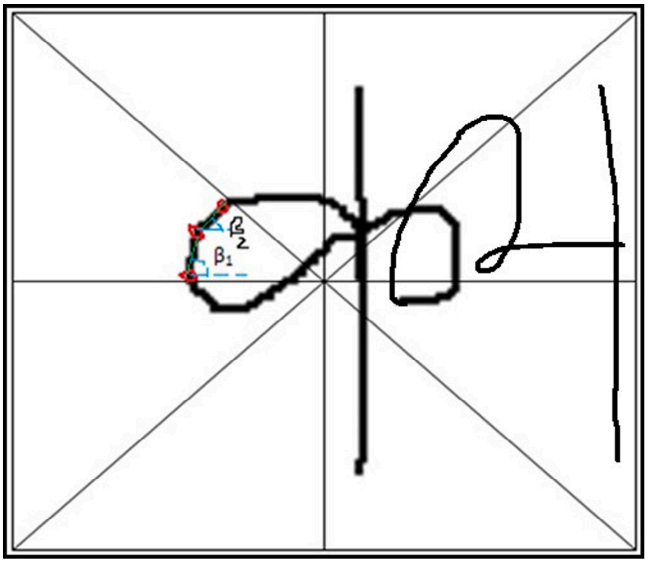

Fig. 8. Illustration of change of trajectory direction feature in one signature in Devanagari script.

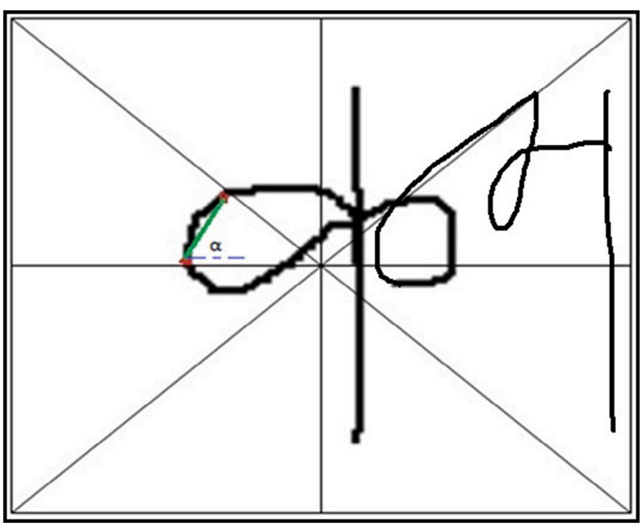

Fig. 9. Illustration of trajectory slope feature in one signature in Devanagari script.

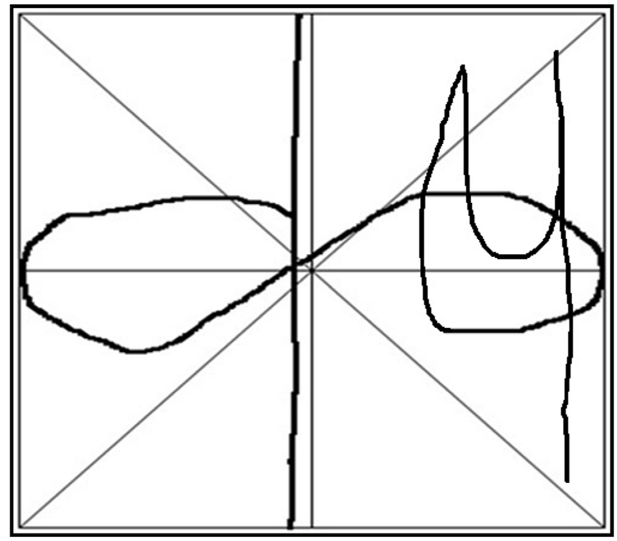

Fig. 10. Illustration of the bounding box and octant division of one Devanagari signature.

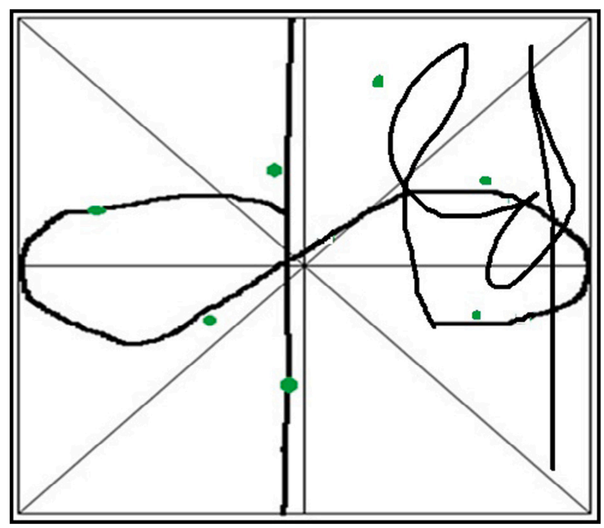

Fig. 11. Centre-of-mass in each octant, denoted by green coloured dot, in one signature sample in Devanagari script.

two pixels is formed which creates an angle α with the x-axis. Cos(α) is considered as the slope of the signature portion available in each octant. Here, Cos(α) provides normalized feature values between -1 and $+1$. If any of the octants does not contain any black pixel then this feature value has been considered as "0" in that octant. Thus, a total of 8 feature values are obtained from this feature. Fig. 9 illustrates the idea of extracting this feature from one signature sample in Devanagari script.

c. Trajectory waviness (TW): To compute this feature, initially, the entire signature image is confined within a bounding box and divided into 8 octants. The feature values are extracted sequentially octant wise in the order from octant 1 to octant 8. This feature is calculated by dividing the Euclidean distance of the signature portion (consists of black pixels) present in each octant by the length of the maximum side (between horizontal and vertical sides) of the octant. The obtained feature values have been scaled down and normalized between 0 and +2. Similarly like earlier two features discussed, If there is no black pixel in any of the octants then this feature value has been considered as "0" in that octant. So, a single feature value is obtained from each octant. Fig. 10 presents the bounding box and octant division of one Devanagari signature.

d. Centre-of-mass (COM): To compute this feature also, the entire signature image is enclosed within a bounding box and divided into 8 octants. The feature values are extracted sequentially octant wise in the order from octant 1 to octant 8. Then the x and y-coordinate values of centre-of-mass of all the black pixels in each octant is computed. These two feature values have been considered as "0" in the octant where there is no black pixel. These feature values have been normalized through dividing both x and y-coordinate values by the length of the maximum side (between horizontal and vertical sides) of the octant. So, another 16 feature values are included in the final feature vector. Fig. 11 shows the centre-of-mass computed in each octant in one signature sample in Devanagari script. Centre-of-mass in each octant is denoted by green coloured dot.

So, a total of 64-dimensional feature vector is generated for each signature sample from the aforesaid four features. Fig. 12 shows the 64-dimensional feature vector of one signature sample written in Latin script from MCYT-75 dataset. Different combinations of features have been tested in the proposed system. *Bayesian optimization technique* has been used to find the optimal value (8 octants) of bin division of each signature image irrespective of any dataset.

4.3. Signature verification and recognition

RNN has been successfully used as a classifier in various research investigations (Ghosh et al., 2019b; Graves, Liwicki, Fernandez, Bertolami, Bunke, & Schmidhuber, 2009) in the field of pattern recognition. RNNs contain self-connected hidden layer. Due to this recurrent connection the internal state of the network can remember the previous inputs. Two special models of RNN classifier, LSTM and BLSTM, have been used in the present investigation for verification and recognition of handwritten signatures. For the BLSTM implementation, the *theano* toolkit[4] has been used. The training time with this toolkit has varied from dataset to dataset in the present system depending on the total number of training samples in each dataset. For LSTM model, the training time has varied between 3 and 5 min for one experiment, whereas for BLSTM model it has varied between 5 and 7 min per experiment. To perform the signature verification and recognition using RNN, entire signature sample wise class labelling scheme of feature vectors has been considered in the training set, i.e., for the feature vectors of all the samples of an individual signature the same class label has been used in the training set. Only the samples of genuine signatures of all the signers have been considered in the training set. Both LSTM and BLSTM models of RNN have been trained using this training set. During usage of the proposed system in real life applications, if some new individual signatures are registered to the existing system then the classifier needs to be re-trained with the newly registered signatures. So, the proposed system works in WD mode. During verification, the feature vectors of the test samples of genuine and all the samples of forged signatures are tested to verify those are genuine or not. For this purpose, the feature vector of the test sample has been fed to LSTM and BLSTM models of RNN and the probability score of the predicted class of the test signature has been extracted at the output layer of RNN. If this score exceeds a certain threshold (mentioned in Section 5.3), then it has been accepted as genuine signature and the corresponding class label is displayed in order to recognize the signature.

In CNN based verification and recognition also, if new signers are registered to the existing system, then the classifier needs to be re-trained with the newly registered signatures. So, the CNN based system also works in WD mode. For verification and recognition using CNN, the probability score of the predicted class of the test signature has been extracted at the output layer of CNN. If this score exceeds a certain threshold (mentioned in Section 5.4), then it has been accepted as genuine signature and the corresponding class label is displayed in order to recognize the signature.

5. Experimental results and analysis

For the experimentation, genuine and forged signatures have been used from the datasets mentioned in Section 3. The classifiers have been trained using the feature vectors of genuine signature samples only of all the signers and the number of training samples is equal for each individual genuine signature in each dataset. The test set contains all the forged signature samples and all the remaining genuine signature samples not present in the training set. Both genuine and forged signature samples in the test set have been used for signature verification purpose, whereas only genuine signature samples in the test set have been used for the recognition purpose. The experimental protocol has been repeated for 10 times by selecting different sets of genuine signature samples in the training and test set.

[4] http://deeplearning.net/software/theano/

5.1. RNN parameters

In the proposed system, LSTM contains one forward hidden layer, whereas BLSTM contains two separate hidden layers—one for processing the input sequence in forward direction and the other for processing it in backwards. The input layer accepts a 64 dimensional feature vector for both LSTM and BLSTM and the hidden layer in LSTM contains 50 recurrently connected memory blocks, whereas both forward and backward hidden layers in BLSTM contain 50 LSTM memory blocks each. The system has been tested with other numbers of memory blocks also, but the one using 50 memory blocks has provided the best performance. Hyperbolic tangent has been used as block input and output activation function. Sigmoid function has been used as gate activation function. The number of neurons in the output layer has been varied according to the number of individual signatures in various datasets for both signature verification and recognition purposes. For GPDS synthetic dataset, 4000 number of neurons have been placed in the output layer as this dataset contains 4000 individual signatures, whereas 75 neurons have been placed in the output layer for MCYT-75 dataset as it contains 75 individual signatures. Similar rule has been followed for other datasets to set the number of neurons in the output layer. In order to use the proposed system in real life applications, the number of neurons in the output layer need to be adjusted according to the number of individual signatures registered in the system. So, if some new writers are enrolling to the existing system then the number of neurons in the output layer has to be increased from the existing one according to the number of newly enrolled writers. This is the another factor due to which the proposed system works in WD mode. Each neuron in the output layer of RNN has a binary output. Each output neuron has been activated with the softmax activation function. The network is optimized by rmsprop optimization algorithm (Hinton, 2015) and categorical cross-entropy loss function with a learning rate of 0.001.

5.2. CNN parameters

In the present system for the experimentation using CNN, a CNN architecture consists of 3 convolutional layers (conv1, conv2, and conv3) followed by two fully connected layers and one output layer has been used. A filter of size 3×3 has been used in each convolutional layer. Maxpooling operation has been performed after each convolutional layer. Pooling operation has been applied on 3×3 subareas of feature matrix obtained from conv1, on 2×2 subareas of feature matrices obtained from conv2 and conv3. Thus, a 9×9 dimensional feature matrix has been obtained from a 128×128 dimensional signature image after third pooling operation. This feature matrix has been flattened and a 81-dimensional feature vector has been fed as input to the fully connected layers. The number of neurons in the output layer is equal to the number of individual signatures in various datasets. Each output neuron in the output layer has been activated with the softmax activation function.

5.3. Signature verification results using RNN

In signature verification, both genuine and forged signatures in the test set have been tested to verify those are genuine or forged. For this purpose, the probability value of the predicted class of the test signature has been extracted at the output layer of RNN. The experimentation suggests that if this probability value is more than 0.6, then a test signature sample is marked as genuine. Two forms of errors have been recorded during signature verification, namely false acceptance of forged signatures [i.e., false acceptance rate (FAR)] and false rejections of genuine signatures [i.e., false rejection rate (FRR)]. FAR indicates the rate of accepting forged signatures as genuine, whereas FRR indicates the rate of rejection of genuine signatures. To measure FAR, forged signature samples of the test set have been used, whereas the genuine

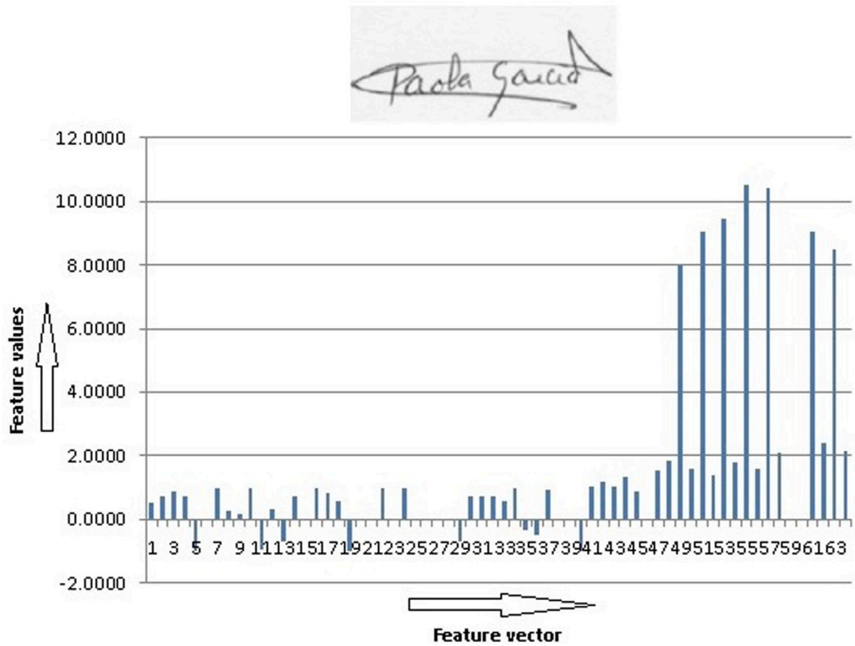

Fig. 12. 64-dimensional feature vector of one signature sample written in Latin script.

Table 2
Signature verification performance in terms of average EER of 10 separate experiments using BLSTM.

Dataset	EER
GPDS synthetic	2.27
GPDS-300	1.46
MCYT-75	0.34
CEDAR	0.01
BHSig260 Hindi	0.43
BHSig260 Bengali	0.36

signature samples of the test set have been used to measure FRR. A trade-off between both of these errors has been established by adjusting a decision threshold. Apart from these two parameters, another metric, known as equal error rate (EER) has also been used to measure the overall error of the verification method. EER is defined as the error rate when FRR = FAR. In signature verification also, the experiment has been repeated for 10 times by selecting different sets of genuine signature samples in the training and test set. Figs. 13 and 14 present the signature verification results of the proposed system in terms of average FAR and FRR obtained from 10 different experiments using BLSTM. Similarly, EER has also been calculated as an average of all the EERs obtained from 10 separate experiments. The detailed statistics of average EER using BLSTM on all the six datasets are presented in Table 2. FAR, FRR, and EER results are presented using BLSTM only because BLSTM model has provided better signature verification performance than LSTM model.

5.4. Signature verification results using CNN

For signature verification using CNN also, the probability value of the predicted class of the test signature has been extracted at the output layer of CNN. The experimentation suggests that if this probability value is more than 0.6, then a test signature sample is marked as genuine. In CNN based verification also, FAR, FRR, and EER have been recorded. To measure FAR, forged signature samples of the test set have been used, whereas the genuine signature samples of the test set have been used to measure FRR. In CNN based signature verification also, the experiment has been repeated for 10 times by selecting different sets of genuine signature samples in the training and test set. So, the EER presented in Table 3 has been computed as an average of all the EERs obtained from 10 different experiments.

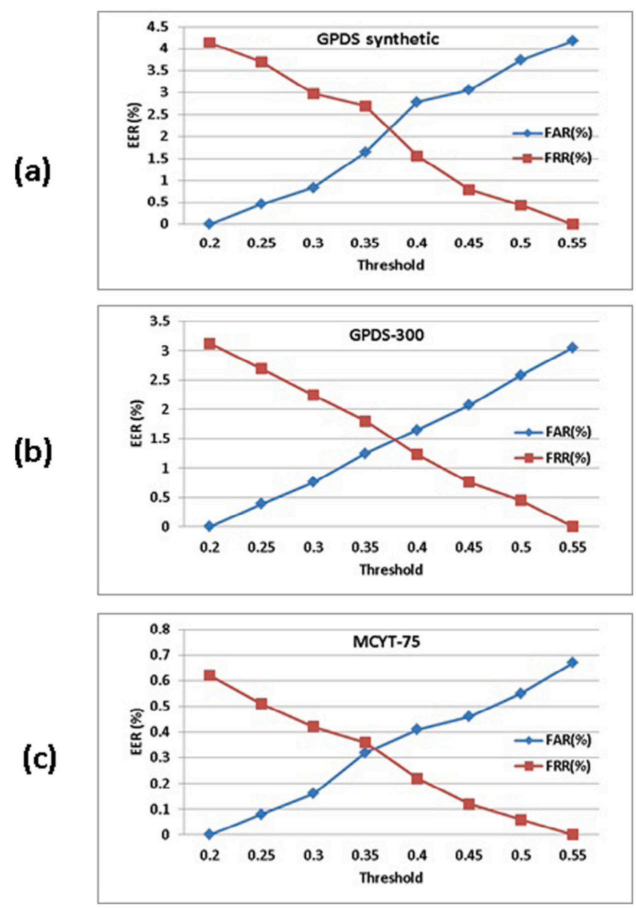

Fig. 13. Signature verification results of the proposed system using BLSTM on: (**a**) GPDS synthetic, (**b**) GPDS-300, and (**c**) MCYT-75 datasets.

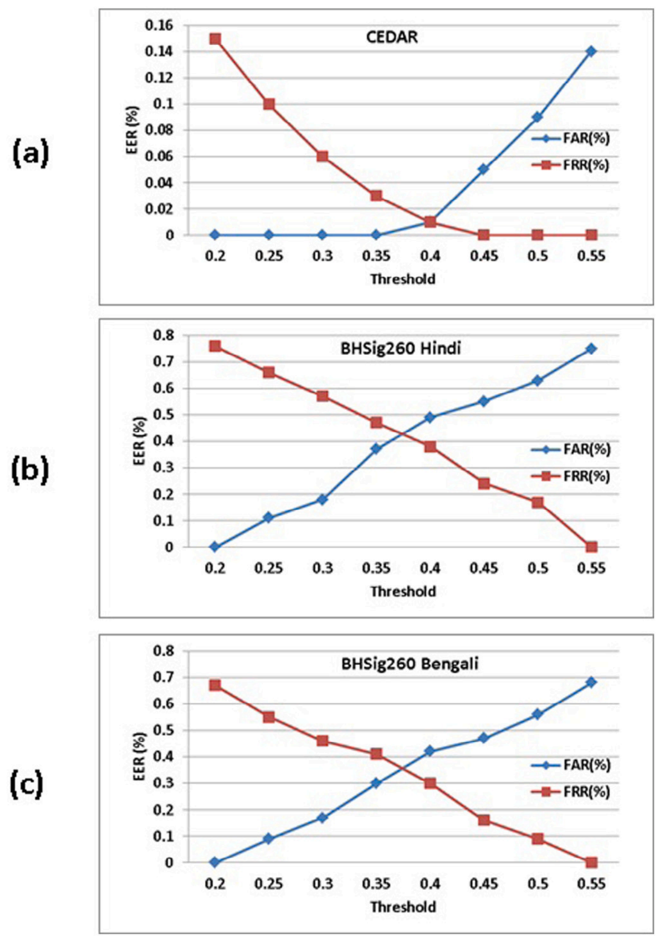

Fig. 14. Signature verification results of the proposed system using BLSTM on: (**a**) CEDAR, (**b**) BHSig260 Hindi, and (**c**) BHSig260 Bengali datasets.

Table 3
Signature verification performance in terms of average EER of 10 different experiments using CNN.

Dataset	EER
GPDS synthetic	5.58
GPDS-300	4.78
MCYT-75	4.54
CEDAR	4.17
BHSig260 Hindi	4.32
BHSig260 Bengali	4.24

Tables 2 and 3 demonstrate that the BLSTM model of RNN performs better than CNN model in verifying the signatures.

5.5. Signature recognition performance using RNN

Here, the signature recognition results using RNN models are presented on all of the six datasets mentioned in Section 3. The experiments have been carried out using both forward LSTM and BLSTM models of RNN with varying number of block size, epoch, and batch size (Ghosh et al., 2019b). "Batch size" indicates the number of input–output set operations performed in each epoch, before updating the weights. The detailed analyses of signature recognition accuracy of the present system are shown in Table 4. Table 4 reports the average recognition accuracy of ten different experiments as the experimental protocol has been repeated for 10 times by selecting different sets of genuine signature samples in the training and test set. It may be noted from this table that low batch size and BLSTM combination has provided higher recognition accuracy.

Table 4
Signature recognition accuracy (RA) of the proposed system.

Dataset	Blocks	Epochs	Batch size	Accuracy	
				LSTM	BLSTM
GPDS synthetic	20	40	24	40.66%	45.60%
	20	40	4	45.18%	51.86%
	50	40	24	76.82%	85.75%
	50	40	4	83.02%	96.08%
GPDS-300	20	40	24	43.18%	46.92%
	20	40	4	46.2%	54.34%
	50	40	24	79.64%	87.43%
	50	40	4	85.76%	98.02%
MCYT-75	20	40	24	45.36%	51.28%
	20	40	4	51.16%	58.24%
	50	40	24	82.37%	90.10%
	50	40	4	91.82%	99.39%
CEDAR	20	40	24	46.82%	53.92%
	20	40	4	52.74%	60.81%
	50	40	24	83.89%	91.63%
	50	40	4	92.13%	99.94%
BHSig260 Hindi	20	40	24	45.94%	53.17%
	20	40	4	52.20%	60.28%
	50	40	24	83.36%	91.11%
	50	40	4	91.61%	99.28%
BHSig260 Bengali	20	40	24	46.08%	53.30%
	20	40	4	52.33%	60.44%
	50	40	24	83.52%	91.29%
	50	40	4	91.79%	99.37%

The results presented in Table 4 demonstrate that the best recognition accuracy is recorded for the combination block size of 50 and batch size of 4 for all of the six datasets. Apart from this, the signature recognition accuracy has also been evaluated by varying the number of features using "Exhaustive Search" feature selection technique for the combination block size of 50 and batch size of 4 using both LSTM and BLSTM. Various feature subsets have been generated using this feature selection technique, the average recognition accuracies of 10 different experiments using some of those subsets are listed in Tables 5 and 6 for LSTM and BLSTM respectively. Some combinations of features in these two tables are showing lower performance due to increase in inter-class similarity from the resultant feature vectors generated from these combinations. Fig. 15 shows different top choices of signature recognition accuracy using both LSTM and BLSTM in all of the six datasets. Here, top three choices i.e., top three probability values during class prediction at the output layer of RNN have been considered.

The performance of the proposed system has also been measured using the Receiver Operating Characteristic (ROC) curve analysis and F1-Score performance measurement parameters. The ROC curve analysis and F1-Score of the proposed system using BLSTM are shown in Fig. 16 and Fig. 17 respectively.

5.6. Signature recognition results using CNN

Signature recognition experiments have also been performed using CNN model. The average recognition accuracy of ten different experiments using CNN model on different datasets is shown in Table 7.

Tables 4 and 7 demonstrate that RNN models perform better than CNN model in recognizing the signatures.

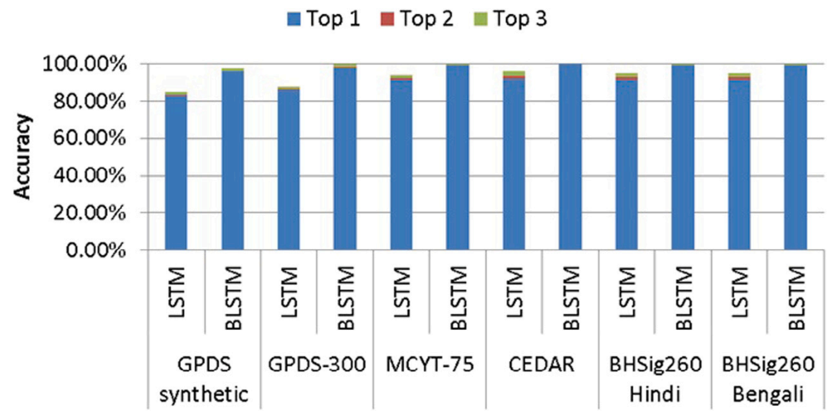

Fig. 15. Signature recognition accuracy of different top choices using LSTM and BLSTM in all six datasets.

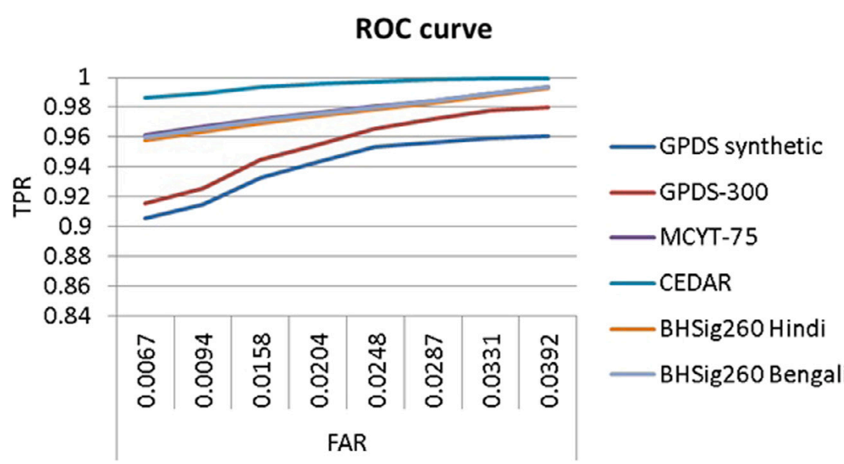

Fig. 16. ROC curve analysis of the proposed system using BLSTM.

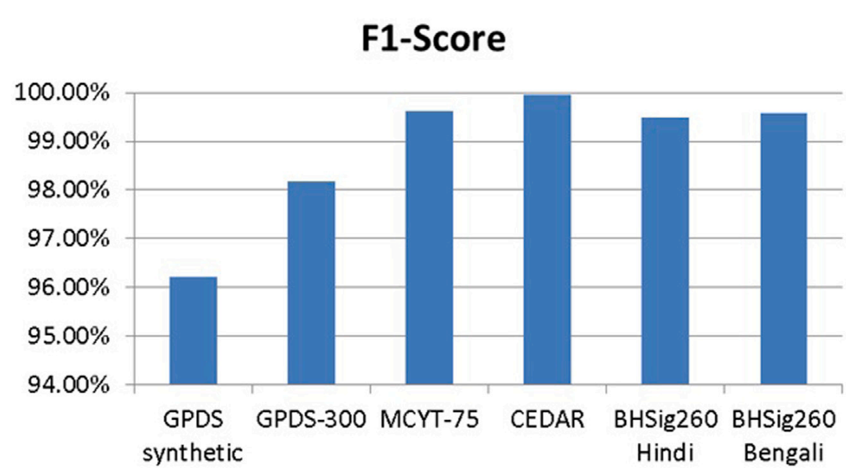

Fig. 17. F1-Score of the proposed system using BLSTM.

Table 5
Signature recognition accuracy using LSTM with block size of 50 and batch size of 4 with varying number of features.

Dataset	Features	Accuracy
GPDS synthetic	CTD, TS, TW, COM	83.02%
	CTD, TS, TW	78.42%
	CTD, TS, COM	74.24%
	CTD, TW, COM	69.32%
GPDS-300	CTD, TS, TW, COM	85.76%
	CTD, TS, TW	81.83%
	CTD, TS, COM	77.79%
	CTD, TW, COM	72.69%
MCYT-75	CTD, TS, TW, COM	91.82%
	CTD, TS, TW	83.78%
	CTD, TS, COM	79.68%
	CTD, TW, COM	74.34%
CEDAR	CTD, TS, TW, COM	92.13%
	CTD, TS, TW	89.34%
	CTD, TS, COM	85.24%
	CTD, TW, COM	80.26%
BHSig260 Hindi	CTD, TS, TW, COM	91.61%
	CTD, TS, TW	87.36%
	CTD, TS, COM	82.43%
	CTD, TW, COM	78.28%
BHSig260 Bengali	CTD, TS, TW, COM	91.79%
	CTD, TS, TW	86.48%
	CTD, TS, COM	82.64%
	CTD, TW, COM	77.34%

Table 6
Signature recognition accuracy using BLSTM with block size of 50 and batch size of 4 with varying number of features.

Dataset	Features	Accuracy
GPDS synthetic	CTD, TS, TW, COM	96.08%
	CTD, TS, TW	89.14%
	CTD, TS, COM	85.32%
	CTD, TW, COM	81.96%
GPDS-300	CTD, TS, TW, COM	98.02%
	CTD, TS, TW	90.92%
	CTD, TS, COM	86.97%
	CTD, TW, COM	83.64%
MCYT-75	CTD, TS, TW, COM	99.39%
	CTD, TS, TW	93.28%
	CTD, TS, COM	89.64%
	CTD, TW, COM	86.42%
CEDAR	CTD, TS, TW, COM	99.94%
	CTD, TS, TW	94.88%
	CTD, TS, COM	91.24%
	CTD, TW, COM	87.96%
BHSig260 Hindi	CTD, TS, TW, COM	99.28%
	CTD, TS, TW	94.32%
	CTD, TS, COM	90.65%
	CTD, TW, COM	87.34%
BHSig260 Bengali	CTD, TS, TW, COM	99.37%
	CTD, TS, TW	94.48%
	CTD, TS, COM	90.80%
	CTD, TW, COM	87.48%

Table 7
Signature recognition accuracy using CNN.

Dataset	Accuracy
GPDS synthetic	93.24%
GPDS-300	94.18%
MCYT-75	94.42%
CEDAR	95.31%
BHSig260 Hindi	95.12%
BHSig260 Bengali	95.19%

5.7. Summarization of the proposed and existing systems

Table 8 presents the summary of some significant signature verification and recognition systems available in the literature and the proposed one. The experimental protocols are not same for all these systems. The verification results are presented in terms of EER, whereas the recognition results are shown in terms of accuracy.

5.8. Strength of the proposed system

1. RNN, through LSTM and BLSTM models, can remember a long sequence of symbols over a longer period of time. Offline signatures are written as a sequence of few known characters in any script or as a combination of few unknown trajectories. In the present system, the feature values of each feature are extracted sequentially octant wise in the order from octant 1 to octant 8 and each octant contains any known character/character portion or unknown trajectory/trajectory portion of the signature, if not vacant. The major strength of the proposed system is it can remember this sequence of characters or unknown trajectories of the signature over a longer period of time, which in turn helps to recognize any signature correctly. This capability of RNN eventually leads to higher signature recognition accuracy of the proposed system in comparison to other ML based signature recognition and verification systems (may be seen in Table 8).
2. BLSTM can remember any sequence in both forward and backward directions. By exploiting this capability of BLSTM, the proposed system can recognize any character or trajectory in a signature by knowing the characters or trajectories appearing both before and after it. This capability is not present in other ML based systems and it eventually leads to higher signature recognition and verification accuracy of the present system in comparison to other ML based systems (may be seen in Table 8).
3. The performance of the proposed system has been tested on six widely used public signature datasets covering signatures from three different scripts—Latin, Devanagari, and Bengali.

6. Conclusion and future work

The present investigation proposes a RNN based deep learning approach towards recognition and verification of offline handwritten signatures. The performance of the proposed system has been evaluated on six widely used public signature datasets and it outperforms existing state-of-the-art methods in this regard. The proposed system also overcomes different drawbacks of shallow ML based signature recognition and verification systems. It is expected that the proposed system will provide fresh acumen to the researchers in developing offline signature recognition and verification systems in other scripts.

In future, it is intended to carry out the investigation in this problem area by including other features as well as employing other deep learning techniques. It is also intended to develop a signature database of people having neurological disorders apart from the signatures of individuals without these disorders and explore the signature verification problem using this dataset. Among other possible directions, it is planned to explore the problem of prediction of Parkinson's disease from handwritten signatures.

Table 8
Summary of some significant signature verification and recognition systems and the proposed one. The experimental protocols are not same for all these systems. Here, NA denotes "Not Available" and TSW indicates "Training Samples per Writer".

Study	Features	Classifier	Dataset	TSW	WD/WI	Performance
Bhunia et al. (2019)	Texture features	SVM	MCYT-75	8	WD	EER: 6.10, RA: NA
			CEDAR	8	WD	EER: 1.64, RA: NA
Chen and Srihari (2006)	Graph Matching	Non-ML	CEDAR	16	NA	EER: 7.9, RA: NA
Diaz et al. (2017)	Textural features	SVM	GPDS-300	8	WD	EER: 14.58, RA: NA
			MCYT-75	8	WD	EER: 9.12, RA: NA
Ferrer et al. (2012)	Grey level features	SVM	MCYT-75	10	NA	EER: 8.16, RA: NA
Guerbai et al. (2015)	Curvelet transform features	SVM	GPDS-300	12	WI	EER: 15.07, RA: NA
			CEDAR	12	WI	EER: 5.60, RA: NA
Hafemann et al. (2018)	Deep CNN features	SVM	MCYT-75	10	WD	EER: 3.40, RA: NA
			CEDAR	10	WD	EER: 2.33, RA: NA
Hafemann et al. (2017)	Deep CNN features	SVM	GPDS-300	12	WI	EER: 1.69, RA: NA
			MCYT-75	10	WI	EER: 2.87, RA: NA
			CEDAR	10	WI	EER: 4.63, RA: NA
Hafemann et al. (2016)	Deep CNN features	SVM	GPDS-300	14	WD	EER: 12.83, RA: NA
Ooi et al. (2016)	DRT+PCA	PNN	MCYT-75	10	NA	EER: 9.87, RA: NA
Pal et al. (2016)	Texture features	KNN	BHSig260 Hindi	12	NA	EER: 24.47, RA: 75.53%
			BHSig260 Bengali	12	NA	EER: 33.82, RA: 66.18%
Serdouk et al. (2016)	Gradient and topological features	AIS	CEDAR	16	WD	EER: 3.54,RA: NA
Soleimani et al. (2016)	Histogram of Oriented Gradients	DMML	GPDS synthetic	10	WI	EER: 20.94, RA: NA
			MCYT-75	10	WI	EER: 9.86, RA: NA
Zois et al. (2020)	Features of image visibility graphs	Non-ML	MCYT-75	10	WD	EER: 1.54, RA: NA
			CEDAR	10	WD	EER: 0.51, RA: NA
Zois et al. (2019a)	Sparse representation based features	SVM	GPDS-300	12	WD	EER: 0.70, RA: NA
			MCYT-75	10	WD	EER: 1.37, RA: NA
			CEDAR	10	WD	EER: 0.79, RA: NA
Zois et al. (2016)	Poset-oriented grid features	SVM	GPDS-300	12	WD	EER: 3.24, RA: NA
			MCYT-75	12	WD	EER: 4.01, RA: NA
			CEDAR	12	WD	EER: 3.02, RA: NA
Proposed method	Structural and directional features	RNN	GPDS synthetic	12	WD	EER: 2.27, RA: 96.08%
			GPDS-300	12	WD	EER: 1.46, RA: 98.02%
			MCYT-75	10	WD	EER: 0.34, RA: 99.39%
			CEDAR	12	WD	EER: 0.01, RA: 99.94%
			BHSig260 Hindi	12	WD	EER: 0.43, RA: 99.28%
			BHSig260 Bengali	12	WD	EER: 0.36, RA: 99.37%

CRediT authorship contribution statement

Rajib Ghosh: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing original draft, and Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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